Homework 1 : Divide and Conquer

Problem 1 1(a).

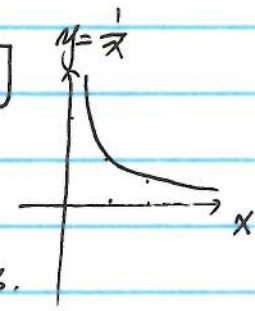
$$T(n) = 2T(n/2) + O(n\lg n)$$
$$= 4T(\tfrac{n}{4}) + 2O(\tfrac{n}{2}\lg\tfrac{n}{2}) + O(n\lg n)$$
$$= 2^2T(\tfrac{n}{2^2}) + O(n\cdot(\lg n - 1 + \lg n))$$
$$= 2^3T(\tfrac{n}{2^3}) + 2^2O(\tfrac{n}{2^2}\lg\tfrac{n}{2^2}) + O(n\cdot(\lg n-1+\lg n))$$
$$= 2^3T(\tfrac{n}{2^3}) + O(n\cdot(\lg n-2+\lg n-1+\lg n))$$
$$\vdots$$
$$= 2^{\lg n}T(1) + O(n\cdot(1+2+\cdots+\lg n-1+\lg n))$$
$$= nT(1) + O(n\cdot\frac{(1+\lg n)\lg n}{2}) = O(n\lg^2 n)$$

$$\left.\begin{array}{l}
T(n) \\
T(\tfrac{n}{2}) \quad T(\tfrac{n}{2}) \to O(n\lg n) \\
T(\tfrac{n}{4}) \quad T(\tfrac{n}{4}) \to 2\cdot O(\tfrac{n}{2}\lg\tfrac{n}{2}) = O(n(\lg n-1)) \\
\vdots \\
T(1) \to 2^{\lg n-1}\cdot O(2\cdot\lg 2) = O(n\cdot 1)
\end{array}\right\} \lg n.$$

$$O(n\lg n) + O(n\cdot(\lg n-1)) + \cdots + O(n\cdot 1) = O(n\lg^2 n)$$

1(b).

$$T(n) = 2\cdot T(n/2) + O(n/\lg n)$$
$$= 2(2\cdot T(n/2^2) + O(\tfrac{n}{2}/\lg\tfrac{n}{2})) + O(n/\lg n)$$
$$= 2^2T(\tfrac{n}{2^2}) + O(n\cdot(\frac{1}{\lg n-1} + \frac{1}{\lg n})]$$
$$\vdots$$
$$= 2^{\lg n}T(1) + O(n\cdot(1+\tfrac{1}{2}+\cdots+\frac{1}{\lg n})]$$
$$= nT(1) + O(n\cdot(1+\tfrac{1}{2}+\cdots+\frac{1}{\lg n})]$$
$$= O[n(1+\tfrac{1}{2}+\cdots+\frac{1}{\lg n})]$$



$\because$ $f(x)=\frac{1}{x}$ is decreasing in $(0,+\infty)$, that is.

$$\int_i^{i+1}\frac{1}{x}dx < \frac{1}{i} < \int_{i-1}^{i}\frac{1}{x}dx.$$ take $i=1,2,\cdots,K$.

sum $\Rightarrow$

$$\int_1^K\frac{1}{x}dx \le \sum_{i=1}^{K}\frac{1}{i} \le \int_1^{K-1}\frac{1}{x}dx+1$$
$$\lg K\cdot\lg e \le \sum_{i=1}^{K}\frac{1}{i} \le \lg(K-1)\cdot\lg e+1$$

$$\left.\begin{array}{l}
T(n) \\
T(\tfrac{n}{2}) \quad T(\tfrac{n}{2}) \to O(n/\lg n) \\
T(\tfrac{n}{4}) \quad T(\tfrac{n}{4}) \to O(n/\lg n-1) \\
\vdots \\
T(1) \to O(n/1)
\end{array}\right\} \lg$$

Thus, $O(n\cdot \lg\lg n) \leq T(n) = O[n\cdot(1+\frac{1}{2}+\dots+\frac{1}{\lg n})] \leq O[n\cdot \lg(\lg n-1)\cdot \lg e+1]$

$\Rightarrow \quad O(n\cdot\lg\lg n) \leq T(n) \leq O(n\cdot\lg\lg n)$

$\therefore T(n) = O(n\cdot \lg\lg n)$

1(c). $T(n) = \sqrt{n}\cdot T(\sqrt{n}) + O(n)$

Assume $n = 2^k$

$T(n) = 2^{\frac{k}{2}}\cdot T(2^{\frac{k}{2}}) + O(2^k)$

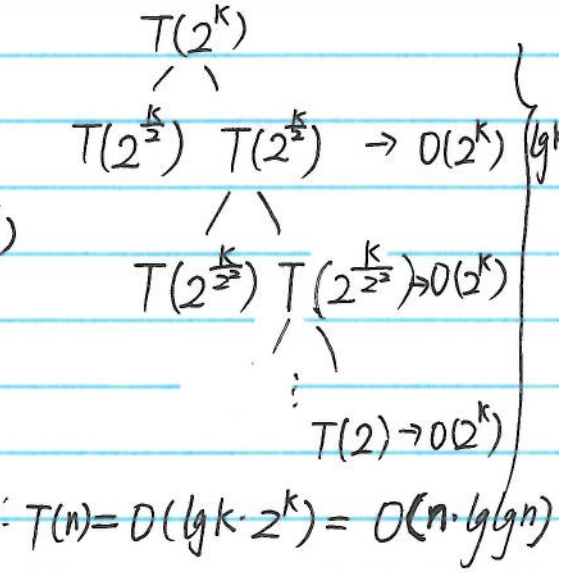
$= 2^{\frac{k}{2}}\cdot[2^{\frac{k}{4}}\cdot T(2^{\frac{k}{4}}) + O(2^{\frac{k}{2}})] + O(2^k)$

$= 2^{k(\frac{1}{2}+\frac{1}{4})}\cdot T(2^{\frac{k}{4}}) + 2O(2^k)$

$\vdots$

$= 2^{k(\frac{1}{2}+\frac{1}{2^2}+\dots+\frac{1}{k})}\cdot T(2) + \lg k\cdot O(2^k)$

$= O(n\cdot\lg\lg n)$



$\therefore T(n) = O(\lg k\cdot 2^k) = O(n\cdot\lg\lg n)$

1(d). $T(n) = T(n/4) + T(n/2) + O(n)$

$= T(\frac{n}{2}) + T(\frac{n}{4}) + O(n)$

$= T(\frac{n}{4}) + T(\frac{n}{8}) + T(\frac{n}{8}) + T(\frac{n}{16}) + O(\frac{n}{2}) + O(\frac{n}{4}) + O(n)$

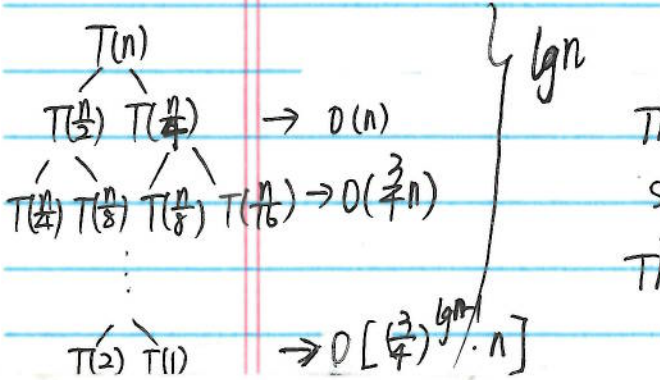
$= T(\frac{n}{4}) + 2\cdot T(\frac{n}{8}) + T(\frac{n}{16}) + O(\frac{3}{4}n) + O(n)$

$= T(\frac{n}{8}) + T(\frac{n}{16}) + 2\cdot T(\frac{n}{16}) + 2\cdot T(\frac{n}{32}) + T(\frac{n}{32}) + T(\frac{n}{64}) +$

$\underbrace{O(\frac{n}{4}) + 2\cdot O(\frac{n}{8}) + O(\frac{n}{16})}_{O((\frac{3}{4})^2 n)} + O(\frac{3}{4}n) + O(n)$

$= T(\frac{n}{8}) + 3\cdot T(\frac{n}{16}) + 3\cdot T(\frac{n}{32}) + T(\frac{n}{64}) + O[(\frac{3}{4})^2\cdot n] + O(\frac{3}{4}n) + O(n)$

$= \dots = C_1T(1) + C_2T(2) + O\left(\sum_{k=0}^{\lg n-1}(\frac{3}{4})^k\cdot n\right] = C_1T(1) + C_2T(2) + O(n)$



There are at most $2^{\lg n} = n$ leaves.

so $C_1T(1) + C_2T(2) \leq O(n)$

Thus, $T(n) = T(\frac{n}{2}) + T(\frac{n}{4}) + O(n)$

$= O(n)$

Problem 2 (a) $M_n(w) = \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & w & w^2 & \cdots & w^{n-1} \\ 1 & w^2 & w^4 & \cdots & w^{2(n-1)} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & w^j & w^{2j} & \cdots & w^{(n-1)j} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & w^{(n-1)} & w^{2(n-1)} & \cdots & w^{(n-1)(n-1)} \end{bmatrix} \begin{array}{l} \leftarrow \text{row for } w^0=1 \\ \leftarrow w \\ \leftarrow w^2 \\ \vdots \\ \leftarrow w^j \\ \leftarrow \\ \quad w^{n-1} \end{array}$

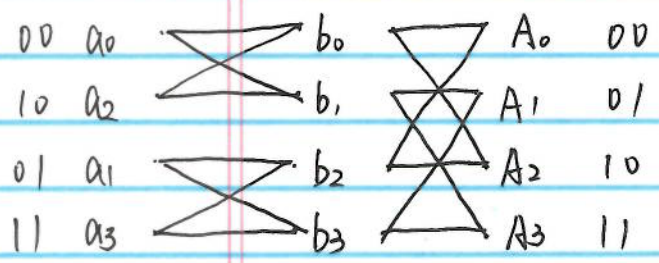
suppose $w^k=1$, $1 \leq k < n$, then the kth row of $M_n(w)$ is
$w^{k-1}, w^{2(k-1)}, \cdots w^{(n-1)(k-1)}$

since $w^n=1$, that means the kth row is exactly the same with the nth row of $M_n(w)$

$\Rightarrow$ that is, $M_n(w)$ has two rows which are linearly dependent

So $M_n(w)$ is not invertible.

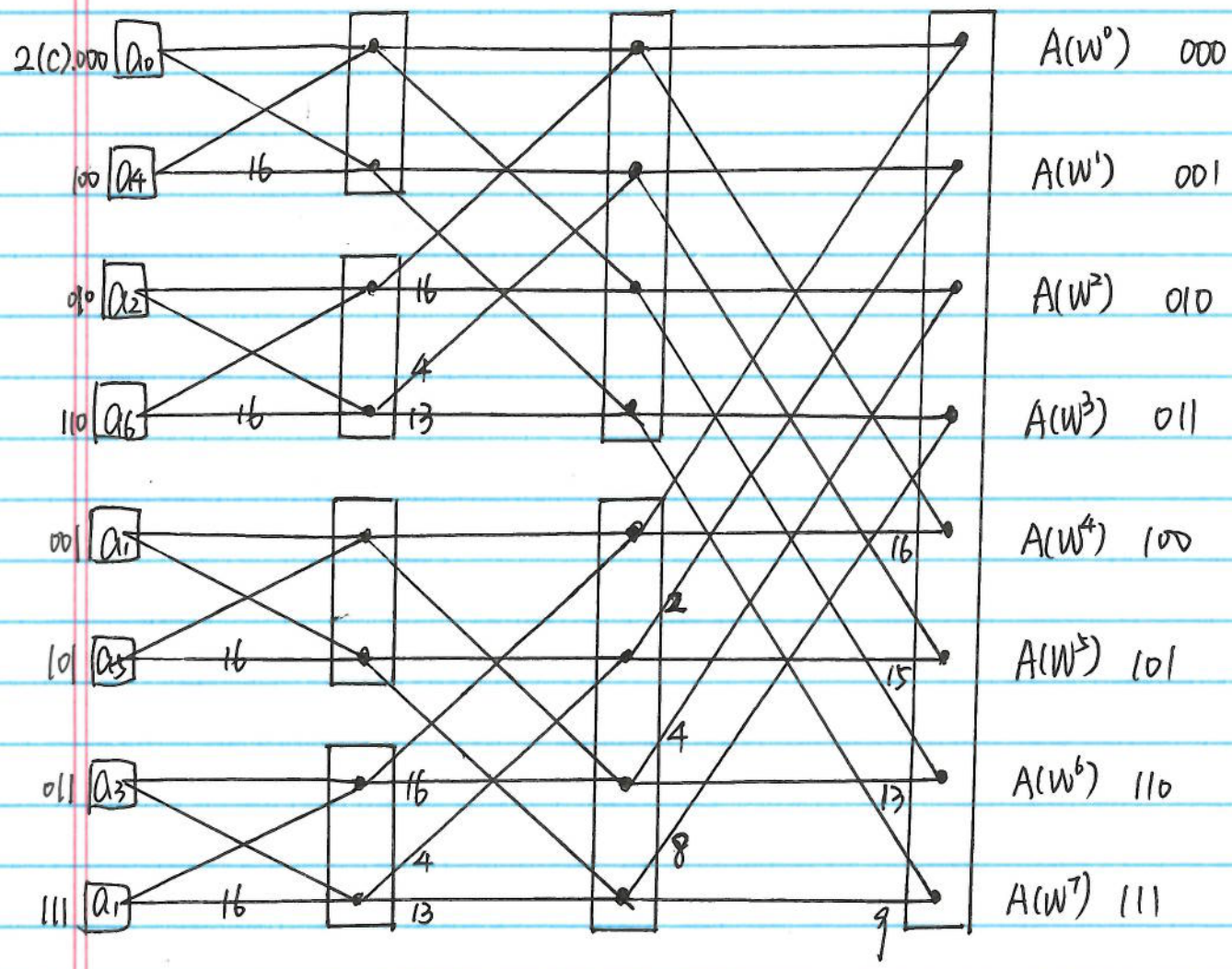
(b) when $n=4$ and $w=i$, $M_n(w) = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{bmatrix}$



$\therefore M_n(w) = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & i \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -i \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & -1 \end{bmatrix}$

$\therefore \begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & -1 \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} b_0 \\ b_1 \\ b_2 \\ b_3 \end{pmatrix}$

$\begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & i \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -i \end{pmatrix} \begin{pmatrix} b_0 \\ b_1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} A_0 \\ A_1 \\ A_2 \\ A_3 \end{pmatrix}$


2(c).000 a0
100 a4
010 a2
110 a6
001 a1
101 a5
011 a3
111 a1
16
16
16
4
13
16
2
4
8
16
15
13
9
A(W^0) 000
A(W^1) 001
A(W^2) 010
A(W^3) 011
A(W^4) 100
A(W^5) 101
A(W^6) 110
A(W^7) 111

Problem 3

Input: • n variable $X=\{x_1, x_2, \dots x_n\}$ of boolean.

• A clauses set $C=\{C_1, C_2, \dots C_m\}$, each clause C_i involves 3 variables whose numbers are within ± 10 of each other.

Output: True or False

Goal: In $O(n)$ time, we want to find an assignment of $X=\{x_1, \dots x_n\}$, which makes the entire formula C evaluate to true.

Dynamic → Programming

To define a subproblem, for each clause C_i, $1 \le i \le m$ (we suppose that for each x_i, it only appears once in one clause, that is C_i; this doesn't matter since we can also hash the C_i into the same group if they all begin with x_i), we try to find J, $J \subseteq \{x_i, x_{i+1}, \dots x_{i+10}\}$ (There are 2^{11} choices for subset J), which make C_i evaluate true.

Remark. if we cannot find such a J, then the answer does not exist, return false

If we can solve each subproblem in $O(1)$ time, the we can finish the whole problem in linear time $O(n)$, cause there are m clauses and $m = O(n)$

fix clause C_i, $1 \le i \le n-10$, suppose we have found such a J, all all 2^{11} choices of $J \subseteq \{x_i, x_{i+1}, \dots x_{i+10}\}$

Now we want to find J' for clause C_{i+1} in $O(1)$ time, where $J' \subseteq \{x_{i+1}, \dots x_{i+11}\}$.

let $J_0 = J' - \{x_{i+11}\}$, $J_1 = J_0 \cup \{x_i\}$, so J_0 and J_1 are subset of $\{x_i, \dots x_{i+11}\}$ that agree with J' on $\{x_{i+1}, \dots x_{i+10}\}$

<u>Case 1</u>: $x_{i+11} \notin J'$: then $J' \subseteq \{x_{i+1}, x_{i+2}, \dots x_{i+10}\}$. we can find assignment, move to nex

<u>Case 2</u>: $x_{i+11} \in J'$ but we can't find an assignment that satisfy C_{i+1}, return false;

<u>Case 3</u>: otherwise. we can find an $J' \subseteq \{x_{i+1}, \dots x_{i+11}\}$, that satisfy C_{i+1}, move to nex

So, we can find J' in $O(1)$ time,

Thus, we can solve the Linear 3SAT in linear time by using dynamic programming.

Problem 4. $A(x) = (x-r_1)(x-r_2)\cdots(x-r_n)$ can be written as $A_L(x)\cdot A_r(x)$.

$A_L(x) = (x-r_1)\cdots(x-r_{\frac{n}{2}})$

$A_R(x) = (x-r_{\frac{n}{2}+1})\cdots(x-r_n)$

Then we can get $A_L(x)$ and $A_R(x)$ by recursion, and we can compute $A_L \cdot A_R$ in $O(\frac{n}{2}\lg\frac{n}{2})$ time. (According to the book, we can compute the product of two degree-d polynomials in $O(d\lg d)$ time)

$$\therefore\ T(n) = 2T(\tfrac{n}{2}) + O(\tfrac{n}{2}\lg\tfrac{n}{2})$$
$$= 2^2T(\tfrac{n}{2^2}) + O(\tfrac{n}{2}\cdot\lg\tfrac{n}{2^2}) + O(\tfrac{n}{2}\lg\tfrac{n}{2})$$
$$= 2^2T(\tfrac{n}{2^2}) + O(\tfrac{n}{2}(\lg n-2 + \lg n-1))$$
$$\vdots$$
$$= nT(1) + O(\tfrac{n}{2}(1+2+\cdots+\lg n-1))$$
$$= O(n\lg^2 n)$$

Problem 5 Input: $A(x) = a_0 + a_1x + a_2x^2 + \cdots + a_{n-1}x^{n-1}$, and n complex numbers $w_0, w_1, \ldots w_{n-1}$.

Output: the n complex numbers $A(w_0), A(w_1), \ldots A(w_{n-1})$.

According to the reference, ~~the~~ we can solve the problem in a time bound of ~~the~~ $O(n\log^2 n)$, that is $c=2$

citations: 1. "Fast Multipoint Evaluation On n Arbitrary Points" by Justine Gauthier., page 25

2. "cs.stackexchange.com/questions/60239/multi-point-evaluation-of-a-polynomial-mod-p"